Reimagining an Abandoned Coastal Social Space Through Community-Driven Speculative Design and Digital Small-Data Methods: A Case Study of a Post-Tourist Hotel in Montenegro

Across post-tourist regions in Southern Europe, abandoned coastal buildings remain striking visual markers yet overlooked in conversations about urban futures, despite evidence that neglected spaces carry distinct social impacts and community value (Escolà-Gascón et al., 2024). This project explores how such spaces might be reimagined not through top-down redevelopment but through participatory, small-scale digital methodologies that foreground community voices, localized knowledge, and emerging cultural practices. The case is an unused seaside hotel on the Montenegrin coast — a region shaped by uneven post-Yugoslav privatization of socialist-era tourist infrastructure, seasonal hyper-tourism, and a post-2022 wave of emigration reconfiguring local social ecologies. The hotel sits at the edge of a thriving summer beach while the building stays largely inactive — yet it has become an informal stage for street art and temporary practices, much as other 'forgotten spaces' act as cultural catalysts rather than physical ruins (Žolobaničová, Štěpánková & Tóth, 2025).

Situated at the intersection of digital humanities, social sciences, and urban studies, the project combines qualitative interviewing, small-data analysis, and speculative 3D modeling to shape a design proposal for a future multifunctional cultural space. Theoretically, it draws on DH traditions of close-and-distant reading and on critiques of scale, which insist computational methods can — and often should — be turned toward small, situated, multilingual corpora. DH thus connects spatial imagination with lived experience and community-oriented design, in line with a wider shift toward data-supported reuse of underutilized structures (Duan et al., 2022). Where Duan et al. mobilize big data on large comment corpora, I adapt their reuse logic to a small, local scale — using deep qualitative data to capture place attachment, seasonal variation, and community practices specific to the Montenegrin coast.

Stage 1: Twenty short on-site interviews were conducted with three groups — Locals (9), Expats (7), Seasonal visitors (4) — a sample consistent with saturation evidence in focused qualitative work (Hennink & Kaiser, 2022). Audio was transcribed locally with Whisper (open-source ASR), keeping participant data on-device — a privacy-preserving choice in a small community where speakers are easily identifiable. Transcripts were coded in two passes: Anthropic's Claude first generated 108 candidate codes, which the researcher consolidated into 37 thematically coherent codes; a human coder then verified and re-applied these in Taguette. Inter-rater agreement reached Cohen's $\kappa = 0.87$ ('almost perfect'), yielding a codebook of 40 codes, 326 highlights, and 417 tag assignments, organized by the Affective / Behavioral / Cognitive framework (Konduri & Lee, 2023) and developed through reflexive thematic logic (Braun & Clarke, 2006; Naeem et al., 2023). Corpus patterns were examined with Voyant Tools and Python scripts. The headline finding is that the three groups hold sharply divergent place imaginaries: Locals foreground regret, loss, and corruption; Expats emphasize nature, access, and infrastructure; Seasonal visitors respond with fascination, art, and creative reuse — an asymmetry informal publics often impose on urban meaning-making before planning frameworks acknowledge it (Garau, Askarizad & Pinna, 2025).

Stage 2 deepens and stress-tests these hypotheses through in-depth semi-structured interviews with several pre-identified informants — among them a local artist who paints on the building, a long-term female resident, and someone who tried to organize events there. The same Whisper — Claude — Taguette pipeline preserves methodological consistency. Stage 3 treats speculative 3D modeling as an analytical instrument, not a presentational artefact: 360-degree photography is processed through Kuula AI to reconstruct the building's current state as spatial evidence read alongside the interview corpus. Luma AI then inserts virtual objects and conceptual elements, letting each interview-derived theme be tested as a spatial hypothesis — asking whether a place imaginary actually fits the volumes, sightlines, and seasonal rhythms of the space. In Stage 4, several concepts are modeled in parallel and circulated with a targeted community survey; results drive final concept selection and the written paper, echoing calls to surface hidden values in underutilized sites (Grazuleviciute-Vileniske & Zmejauskaite, 2025).

I argue this approach — bottom-up engagement, small-data analysis with verifiable AI-assisted coding, and speculative design used analytically — offers a meaningful alternative to both large-scale redevelopment and purely ecological reinterpretation of abandoned structures. Treating the hotel not as a problem but as a social catalyst, the project reframes urban transformation as an engaged, iterative, collaborative act. The poster will present the complete research arc — from the interview corpus and codebook through the 3D analytical model and community survey to the final design concept — demonstrating how DH methods can support inclusive imagining of future use for spaces that exist outside heritage or development narratives yet remain woven into everyday coastal life.

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